

Moksh Shah

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EDUCATION

Toronto Metropolitan University

Sept 2021 – May 2026

BSc (Hons), Computer Science – Software Engineering Concentration

Toronto, ON

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Relational Databases, Object-Oriented Programming, Functional Programming, Software Development Life Cycle

EXPERIENCE

SEW-Eurodrive Canada

Nov 2025 – Present

Software Developer

Toronto, ON

- Architected a custom **Linux API** to capture, process, and persist **ROS2** log messages into a **SQL** database, enabling centralized logging infrastructure across distributed robotics systems.
- Built a web-based analytics **dashboard** with **Python** and **Flask** delivering real-time log filtering by error severity, timestamp, and multi-parameter sorting.
- Designed a specialized **LLM** pipeline for **code analysis** across repositories totaling **100,000+ lines**, achieving **sub-1-second query response times** for implementation guidance.
- Implemented a real-time dynamic obstacle avoidance system in **ROS2**, significantly improving moving-object tracking accuracy over the legacy static detection system.

SOTI

Sept 2023 – June 2024

Software Developer Intern

Mississauga, ON

- Migrated product architecture from sequential **MDM** to parallel **DDM**, boosting response times by **70%** using **C#**, **TypeScript**, **AngularJS**, **HTML**, and **SCSS**.
- Optimized the **CI/CD pipeline** and resolved race conditions in legacy code, reducing build times from **8 hours to 30 minutes (93% reduction)**.
- Authored **unit tests** achieving **90%+ code coverage** using behavior-driven testing frameworks, significantly improving validation reliability.
- Took on junior PM duties: drafted **Jira** tickets, translated client requirements, and coordinated with the dev team.

Toronto Metropolitan University – Autonomous Vehicles Lab

May 2023 – Sept 2023

Research Assistant

Toronto, ON

- Led a team of **5** graduate aerospace engineers to build a **LiDAR-based SLAM** system for autonomous navigation on rovers and drones.
- Deployed **TensorFlow Lite** object detection models on **Raspberry Pi**, optimized via **semantic segmentation** for embedded performance constraints.
- Assembled a research rover with a custom **Linux** distribution using **C++** and **Python**; conducted workshops for **10+** graduate students on **Bash**, **Git**, **ROS2**, **CMake**, and **Gazebo** simulations.

Studio 98

April 2022 – Aug 2022

Software Developer Intern

Clearwater, FL

- Built the front-end framework for a business management app using **TypeScript** and **React Native**, delivering **cross-platform** iOS/Android compatibility and improving development efficiency by **50%**.
- Integrated **REST API** calls for backend data, built a login/form authentication layer, and engineered a modular file structure to enhance scalability and reduce concurrent rewrite risk.

AWARDS

University of Guelph CTF – 3rd Place (Nov 2024) | Splunk CTF – 3rd Place | Dean's List | Scotiabank Career Builder Program

TECHNICAL SKILLS

Languages: Python, C++, C, C#, Java, TypeScript, JavaScript, SQL, Kotlin, R, HTML/CSS, SCSS

Frameworks & Libraries: Node.js, Flask, Django, REST APIs, AngularJS, Vue.js, Next.js, React Native, .NET, ASP.NET, JUnit, Selenium, NumPy, Pandas, OpenGL

Tools & Platforms: Git, Docker, Kubernetes, Jenkins, Linux/Unix, Ubuntu, Kali Linux, Jira, Firebase, AWS-style CI/CD, Wireshark, Figma, Gazebo, ROS2

Concepts: Data Structures & Algorithms, OOP, Distributed Systems, Microservices, Concurrency, CI/CD, Agile/SCRUM, Unit Testing